Hypertension

ORIGINAL ARTICLE

Primary Aldosteronism Medical Treatment Outcomes (PAMO) Beyond 12 Months

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BACKGROUND: Primary aldosteronism (PA) is a common and treatable cause of hypertension, yet data on the durability of medical treatment response beyond the first year are limited. Using the PA medical treatment outcomes framework, we evaluated long-term biochemical and clinical outcomes and factors associated with treatment response.

METHODS: We conducted an international, multicenter, observational cohort study across 27 centers on 4 continents. Adults with PA treated with mineralocorticoid receptor antagonists and epithelial sodium channel blockers between 2016 and 2021 were included if follow-up data were available at ≥12 months. Biochemical and clinical responses were classified as complete, partial, or absent using PA medical treatment outcomes criteria. Ordinal logistic regression identified determinants of a favorable treatment response.

RESULTS: The cohort included 1292 patients (47.5% women, mean age 52 years) with a median follow-up of 34 months (interquartile range, 21–51). At ≥12 months, 58.3% achieved a complete biochemical response, and 20.8% achieved a complete clinical response, with only 5% showing no response. Determinants of a favorable biochemical response included lower aldosterone and lateralization index, and higher renin and potassium levels at baseline. In those taking spironolactone, a higher daily dose was also associated with a favorable biochemical response. Determinants of a favorable clinical response included female sex, lower body mass index, shorter duration of hypertension, and absence of hypertension-mediated organ damage.

CONCLUSIONS: Targeted medical therapy for PA can deliver sustained biochemical and clinical benefits. Early disease detection and adequate dose titration are highly actionable determinants of long-term treatment success. ***(Hypertension.* 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.126.27431.)** • **Supplement Material**.

Key Words: aldosterone ■ blood pressure ■ cardiovascular disease ■ potassium ■ renin

Hypertension is the leading risk factor for cardiovascular disease worldwide, yet control rates remain poor at ≈21%.[1] A common and highly modifiable cause of hypertension, primary aldosteronism (PA), is characterized by dysregulated adrenal aldosterone production independent of renin status.[2] Untreated PA is associated with higher odds of cardiovascular, cerebrovascular, kidney and metabolic diseases compared with primary hypertension.[3,4] These risks can be mitigated through targeted treatment, either by unilateral adrenalectomy in

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ORIGINAL ARTICLE

NOVELTY AND RELEVANCE

What Is New?

The Primary Aldosteronism (PA) Medical treatment Outcomes beyond 12 months study is the first large international analysis to demonstrate that complete biochemical control of aldosterone excess, reflected by potassium and renin normalization, can be sustained beyond 12 months, particularly with an adequate dose of spironolactone (or other mineralocorticoid receptor antagonist). The clinical response is also durable with significant blood pressure reduction in the majority of patients, even if additional antihypertensives are still required. Features suggestive of less severe or earlier disease, such as shorter duration of hypertension or absence of known hypertension mediated organ damage, were also associated with a more complete clinical response.

What Is Relevant?

These findings clarify why blood pressure alone is an incomplete marker of treatment adequacy in PA. Aldosterone excess drives sodium retention, tissue inflammation, fibrosis, and endothelial dysfunction in addition to elevated blood pressure. Renin normalization provides a mechanistic indicator of effective mineralocorticoid receptor blockade beyond blood pressure per se. The greater likelihood of a favorable clinical response in patients with shorter hypertension duration highlights the cumulative nature of aldosterone-mediated tissue injury and supports the biological importance of early diagnosis. The consistency of these associations across a large multinational cohort over a median of almost 3 years supports the generalizability of the underlying pathophysiology and its relevance across health care systems.

Clinical/Pathophysiological Implications?

Together, these data support the use of both renin normalization and blood pressure control as treatment targets in the management of PA. This framework provides a rational basis for mineralocorticoid receptor antagonist titration, particularly when blood pressure appears controlled, but renin remains suppressed, although this approach remains to be tested in randomized controlled trials. Pathophysiologically, the findings reinforce PA as a modifiable disease in which early correction of aldosterone excess can increase the likelihood of optimal clinical response and may even alter long-term cardiovascular risk trajectories. Clinically, these results strengthen the case for earlier detection and timely intervention of PA to prevent irreversible end-organ damage and maximize the likelihood of complete clinical response.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
DDD	defined daily dose
MRA	mineralocorticoid receptor antagonist
OR	odds ratio
PA	primary aldosteronism
PAMO	Primary Aldosteronism Medical treatment Outcomes

patients with lateralizing disease, or with MRA (mineralocorticoid receptor antagonists) in those with bilateral subtypes of PA or who are not surgical candidates.[5–7]

We previously proposed the PA Medical Treatment Outcomes (PAMO) criteria, a standardized framework for categorizing medical treatment responses in PA.[8] In an international cohort of 1258 patients, 52.8% achieved a complete biochemical response, defined by the normalization of serum potassium and plasma renin concentration, by 6 to 12 months after treatment initiation. However, only 18.3% achieved a complete clinical response, defined as blood pressure normalization on targeted treatment alone.

Although a biochemical response can be achieved within 12 months of MRA therapy, it is possible that a full clinical response may take longer due to the time needed for end-organ recovery, such as the reversal of aldosterone-mediated vascular fibrosis and arterial stiffness.[9,10] Furthermore, dose titration of targeted therapy may be limited by long waiting times in some centers,[11] meaning the extent of biochemical and clinical response may not be fully captured within the first year of follow-up. In this study, we therefore aimed to evaluate longer-term biochemical and clinical outcomes in the same international cohort, and to identify factors independently associated with favorable treatment response beyond the initial 6 to 12 months.

METHODS

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design

We conducted an international, multicenter, observational follow-up cohort study, as an extension of the original PAMO study,[8] to evaluate longer-term treatment outcomes in patients with PA.

Study Sites and Participants

All 31 centers from the original PAMO study were invited to contribute patient data for the assessment of medical treatment outcomes at or beyond 12 months of follow-up, of which 27 centers were able to accept. At each participating center, consecutive patients with PA, diagnosed according to the Endocrine Society guideline criteria[12] and local protocols, were eligible for inclusion if they were receiving targeted pharmacological treatment (MRA and epithelial sodium channel inhibitor). These included patients with bilateral (or nonlateralizing in those who had adrenal vein sampling PA, those with unilateral (or lateralizing) PA who opted for medical treatment, those who required medical treatment after incomplete surgical cure, and those with an indeterminate subtype. Medical treatment needed to be initiated between January 2016 and December 2021, with follow-up data for either clinical or biochemical outcomes at or beyond 12 months from initiation. Patients were excluded if they lacked follow-up data at or beyond 12 months, even if they had been included in the original PAMO study with 6-12 months of follow-up.[8]

Data Collection

Data were collected using a standardized template and included: demographic details, baseline office blood pressure, defined daily dose (DDD) of antihypertensive medications, biochemistry, presence of diabetes, microalbuminuria and left ventricular hypertrophy, results of subtyping procedures (where the subtype was defined based on institutional thresholds), and treatment response at least 12 months after initiation of targeted therapy. The validity of data sources was supported by each center's prior track record in research and clinical expertise in the field of PA. Data cleaning, consisting of systematic checking for outliers, internal consistency, and plausibility of key variables (eg, verifying medication dosage against the calculated DDD), was performed centrally by 3 investigators (J.B., J.G., and J.Y.). Queries regarding missing or inconsistent data were resolved in consultation with the respective site investigators. Categorization of patients' biochemical and clinical outcomes as complete, partial, or absent, according to PAMO criteria,[8] was completed by each center, and verified by J.Y., J.B., and J.G.

All participating centers obtained approval from their respective institutional ethics committees for the analysis of deidentified retrospective data. A data sharing agreement was established with each site. Individual patient consent was not required as the study used deidentified retrospective patient data. The overall project was approved by the Human Research Ethics Committee at Monash Health, Australia (RES-23-0000-365Q).

The primary outcome was the proportion of patients classified into each of the 3 PAMO response categories at ≥12 months posttreatment: biochemical and clinical outcomes were assigned using data from the most recent available follow-up visit occurring at or beyond 12 months after initiation of targeted medical therapy (outcomes were assessed at a discrete follow-up time point and did not represent cumulative, average, or best-achieved responses over the follow-up period). Secondary outcomes included the identification of factors associated with a complete biochemical or clinical response.

Statistical Methods

Statistical analyses were performed using IBM SPSS Statistics 29.0 (IBM Corp., Armonk, NY), GraphPad Prism 10.0 (GraphPad Software, San Diego, CA), and R (R Core Team, Vienna, Austria; version 2026.01.0+392), within the R Studio integrated development environment (Posit Software, PBC). Continuous variables were expressed as mean and SD when distribution was symmetrical, or as median with interquartile range when skewed, then the corresponding parametric (Student *t* test or 1-way ANOVA with Bonferroni post hoc test) or nonparametric tests (Mann-Whitney *U* test or Kruskal-Wallis test) were applied according to type of variable, sample size and distributional characteristics, as appropriate. Categorical variables were reported as frequency and percentage between brackets and analyzed by χ^2 or Fisher exact test, as appropriate. Pairwise comparisons across the 3 response categories for biochemical and clinical outcomes were conducted only when the overall group comparison was statistically significant, to identify which specific groups differed. The agreement between biochemical and clinical outcomes was assessed using cross-tabulation and weighted Cohen kappa.

A 3-level proportional odds ordinal logistic regression model was fitted to evaluate baseline parameters associated with biochemical and clinical outcomes, treated as ordered categories (from absent to partial to complete response, with increasing favorable levels of outcome). The proportional odds assumption was formally assessed using the Brant test; the derived adjusted odds ratios (ORs) represent the likelihood of being in a more favorable outcome category. Covariates and potential confounding variables included in the explanatory models were specified a priori based on clinical relevance and current knowledge. *P* values lower than 0.05 were considered significant.

RESULTS

The study included data from 1292 patients across 27 participating centers (Table S1), with a median follow-up of 34 months (interquartile range, 21–51). The mean age at referral was 52 years (SD, 11.3), and 47.5% were women (n=614). Baseline characteristics are summarized in Table S2. Most patients had bilateral PA (n=1035, 80.1%; n=70 for unilateral PA; n=187 for indeterminate subtypes), with significant regional variations in demographics and treatment (Table S3). Geographic area was included as a covariate in multivariable models.

Clinical and biochemical outcomes at ≥12 months following the initiation of targeted medical treatment were evaluated using the PAMO criteria, stratified by center (Figure S1). Among the 1100 participants with paired clinical and biochemical outcomes, 14.4% (n=158) achieved complete biochemical and clinical response, while 5% (n=55) had absent response in both domains (Figure [A]). The most common outcome was complete biochemical response with partial clinical response, observed in 38.6% (n=424) of the participants. An exact agreement between the outcome categories was observed in 367 patients (33.4%), with a significant association between biochemical and clinical response

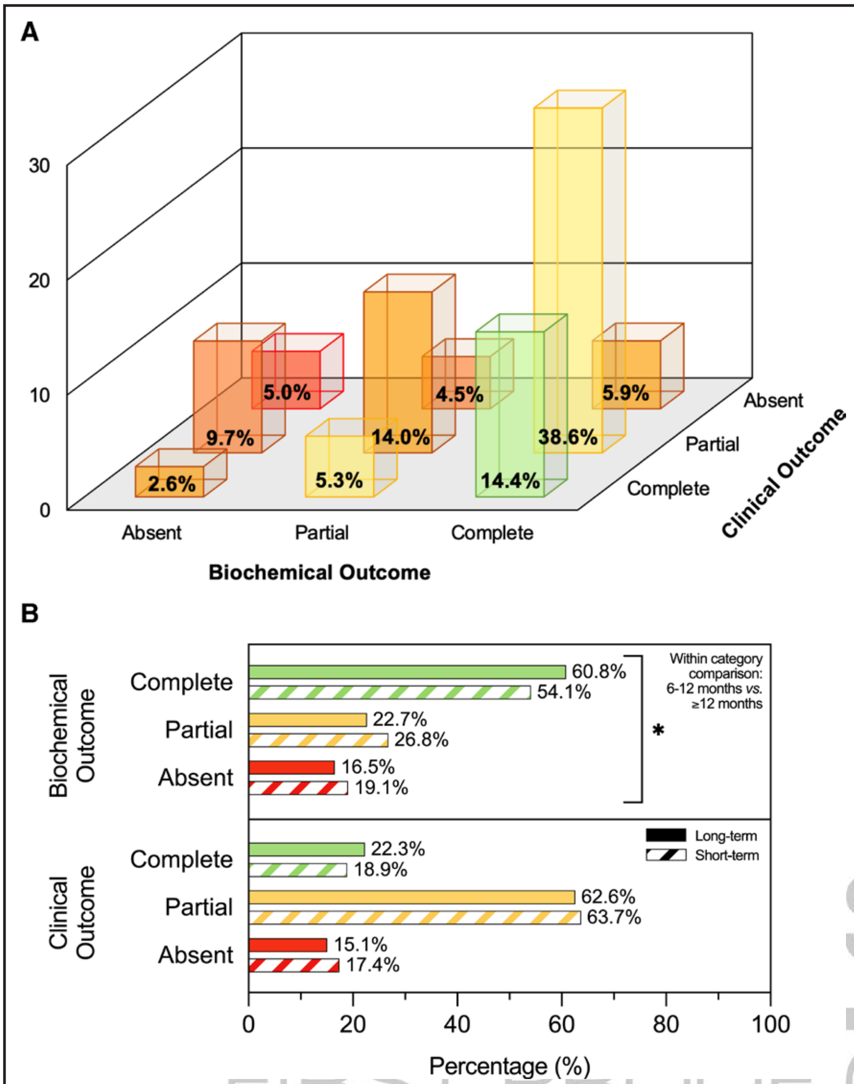

Figure. Biochemical and Clinical Outcomes according to Primary Aldosteronism (PA) Medical treatment Outcomes (PAMO) consensus.
A, Rate of biochemical response stratified for clinical outcome (n=1100). **B**, Rates of biochemical and clinical responses at short- and long-term follow-up. For participants with paired data available at both time points, PAMO response categories at 6–12 months after treatment initiation (short-term) were compared with those at ≥12 months (long-term); the comparison was performed by χ^2 test evaluating changes over time within each response category. Paired data were available for n=868 participants for biochemical response and n=1072 for clinical response. An improvement was observed in patients' biochemical outcomes: the rate of complete biochemical response was higher at long-term follow-up (P=0.004), and there were fewer partial responders (P=0.045). The lower rate of absent biochemical response at long-term did not reach statistical significance (P=0.149). *P<0.05.

(χ^2=44.9; P<0.001); however, weighted Cohen's kappa was 0.03 (P=0.236), indicating low concordance between the response domains. Among the 868 participants with paired follow-up biochemical data at both 6-12 months and ≥12 months, the across-time rate of complete biochemical response was significantly higher at ≥12 months (60.8% versus 54.1%; P=0.004; Figure [B]). A similar but nonsignificant trend was observed in the clinical response.

Among the 1120 participants with follow-up biochemical data, 58.3% (n=653) achieved a complete biochemical response, while 17.6% (n=197) had an absent response (Table S2). Of 1272 participants with follow-up clinical data, 20.8% (n=264) achieved a complete response, and 15.5% had an absent response (n=197). Response rates varied significantly by geographic region (Figure S1; Table S3). The highest rate of complete biochemical response was observed in the Americas (72.4%), followed by Asia (58.7%), Oceania (57.5%), and Europe (52.7%). Similarly, complete clinical responses were most common in the Americas and Oceania (both 30.2%), followed by Asia (17.7%) and Europe (17.1%; Table S3). When stratified by subtype, patients with unilateral PA had a significantly higher rate of absent biochemical response compared with those with bilateral PA (29.2% versus 16.5%; P=0.032), although clinical response categories were similar between subgroups (Table S4).

At baseline, the average age was lower in women (n=614) than in men (n=678; Table S5). Women had a shorter duration of hypertension, lower body mass index, lower systolic and diastolic blood pressure, and lower DDD of antihypertensive medications. They also had higher estimated glomerular filtration rate, and fewer had diabetes or hypertension-mediated end-organ damage. After targeted treatment, a significantly greater proportion of women achieved complete clinical response compared with men (29.3% versus 12.9%; P<0.001) despite receiving a lower DDD of MRA (1.05 versus 1.10; P=0.043). Of note, the composition of MRA was different between women and men—spironolactone represented 82.5% of MRA in women and 62.9% in men, while eplerenone represented 11.4% of MRA in women and 32.6% in men. The magnitude of blood pressure reduction and the proportion achieving a complete biochemical response were comparable between sexes.

Data stratified for biochemical and clinical response are reported in Tables 1 through 4. When stratified for biochemical response, participants with a complete biochemical response (n=653) were similar to those with an absent response (n=197) in their demographic and clinical characteristics at baseline (Table 1). However, the complete responders had a significantly lower baseline aldosterone-to-renin ratio (124 versus 149 pmol/mU) and higher serum potassium concentration (3.9 versus 3.7 mmol/L) compared with the absent responders. After follow-up for a median duration of 34 months (interquartile range, 21–51), the complete biochemical responders displayed significantly lower systolic and diastolic blood pressure compared with the absent responders (systolic, 130 [SD, 14.2] versus 139 [SD,

Table 1. Baseline Characteristics of Patients After Stratification for Biochemical Response

Variable	Biochemical outcome			Overall *P* value	Pairwise comparisons		
	Complete (n=653)	Partial (n=270)	Absent (n=197)		C vs P	C vs A	P vs A
Age at referral, y	52±11.3	51±10.8	52±11.0	0.709	…	…	…
Duration of HTN, mo	72 (13–158)	64 (12–152)	64 (13–145)	0.699	…	…	…
Sex (ref. female), n (%)	300 (45.9)	138 (51.1)	92 (46.7)	0.353	…	…	…
BMI, kg/sqm	29.1±5.86	27.8±5.32	28.3±5.91	0.010*	0.010*	0.375	1.000
SBP, mm Hg	151±19.5	149±21.0	151±18.0	0.426	…	…	…
DBP, mm Hg	92±13.0	91±11.9	91±11.6	0.869	…	…	…
DDD	2.00 (1.00–3.66)	2.00 (1.00–3.00)	2.00 (1.00–3.00)	0.478	…	…	…
Use of βB (ref. yes), n (%)	126/649 (19.4)	47/267 (17.6)	32/193 (16.6)	0.614	…	…	…
AC, pmol/L	533.9 (396.7–747.8)	495.2 (354.3–740.8)	607.4 (424.1–788.8)	0.003*	0.215	0.052	0.002*
PRA, ng/mL per hour	0.45 (0.21–0.60)	0.20 (0.10–0.33)	0.40 (0.20–0.60)	<0.001*	<0.001*	1.000	<0.001*
DRC, mU/L	4.0 (2.0–6.5)	2.0 (1.7–3.3)	3.5 (2.0–5.9)	<0.001*	<0.001*	1.000	<0.001*
ARR (PRA)	1318 (821–2339)	2682 (1183–5450)	1531 (855–3678)	<0.001	<0.001	0.768	0.003*
ARR (DRC)	124 (80–215)	192 (123–377)	149 (96–263)	<0.001*	<0.001*	0.011*	0.005*
Creatinine, mg/dL	0.88±0.272	0.88±0.396	0.90±0.289	0.840	…	…	…
eGFR, mL/min	87±19.4	87±19.6	84±19.7	0.127	…	…	…
K^+, mmol/L	3.9±0.44	3.8±0.43	3.7±0.52	<0.001*	0.004*	<0.001*	0.819
Lowest K^+, mmol/L	3.4±0.51	3.4±0.47	3.2±0.56	<0.001*	0.469	<0.001*	0.020*
Diabetes (ref. yes), n (%)	124 (19.0)	36 (13.3)	27 (13.7)	0.052	…	…	…
MA (ref. yes), n (%)	95/492 (19.3)	39/207 (18.8)	32/154 (20.8)	0.892	…	…	…
LVH at echo (ref. yes), n (%)	172/543 (31.7)	73/221 (33.0)	52/154 (33.8)	0.860	…	…	…
CT scanning							
Nodule (ref. yes), n (%)	202/627 (32.2)	98/260 (37.7)	74/191 (38.7)	0.128	…	…	…
Adrenal description, n (%)				0.128	…	…	…
Bilaterally normal	276/607 (45.4)	106/249 (42.5)	64/186 (34.4)				
Bilaterally abnormal	97/607 (16.0)	41/249 (16.5)	35/186 (18.8)				
Unilateral abnormality	234/607 (38.6)	102/249 (41.0)	87/186 (46.8)				
Adrenal venous sampling without ACTH							
Lateralization index	2.1 (1.3–3.2)	2.3 (1.5–4.5)	2.0 (1.5–5.2)	0.146	…	…	…
CL suppr. (ref. yes), n (%)	110/296 (37.2)	40/102 (39.2)	36/78 (46.2)	0.350	…	…	…
Adrenal venous sampling with ACTH							
Lateralization index	1.6 (1.2–2.5)	1.6 (1.2–2.2)	1.8 (1.3–2.9)	0.392	…	…	…
CL suppr. (ref. yes), n (%)	50/241 (20.7)	24/124 (19.4)	24/90 (26.7)	0.399	…	…	…

Baseline patient characteristics after stratification for biochemical outcome according to PAMO criteria. AC indicates aldosterone concentration; ACTH, adreno-corticotropic hormone; ARR, aldosterone-to-renin ratio; BMI, body mass index; CL, contralateral; CT, computed tomography; DBP, diastolic blood pressure; βB, β-blockers; DDD, defined daily dose; DRC, direct renin concentration; echo, echocardiography; eGFR, estimated glomerular filtration rate; HTN, hypertension; LVH, left ventricular hypertrophy; MA, microalbuminuria; PAMO, Primary Aldosteronism Medical treatment Outcomes; PRA, plasma renin activity; SBP, systolic blood pressure; and suppr., suppression.

*A $P<0.05$ was considered significant.

18.6] mm Hg; diastolic 82 [SD, 10.3] versus 87 [SD, 12.2] mm Hg), as well as a greater decrease in blood pressure and antihypertensive DDD (Table 2). For the majority of participants on spironolactone as targeted therapy (n=815), the median daily dose was significantly higher in the complete biochemical responders (n=520) compared with absent responders (n=117; 40 versus 25 mg; *P*=0.007), although the DDD of all MRA combined (n=1120, including n=249 for eplerenone, n=20 for esaxerenone) was not significantly different across the response categories (Table 2).

In an ordinal logistic regression analysis, factors independently associated with a better biochemical response (complete > partial > absent) included lower baseline aldosterone (OR, 0.999 [0.998–0.999]; *P*=0.035), higher baseline renin (measured as concentration or activity; OR, 1.097 [1.045–1.152]; *P*<0.001), higher baseline potassium concentration (OR, 1.543 [1.065–2.233]; *P*=0.022), and an LI lower than 4 at adrenal vein sampling (OR, 0.666 [0.448–0.989]; *P*=0.044), taking into account of geographic area, follow-up duration, MRA dose and use of β-blocker as potential confounding factors (Table 5).

When stratified by clinical response, participants with a complete clinical response (n=264) differed significantly from those with an absent clinical response (n=197) across several baseline characteristics (Table 3). The complete response group had a lower body mass index (26.9 versus 28.3; *P*=0.048), shorter duration of hypertension (24 versus 96 months; *P*<0.001), lower antihypertensive requirement (DDD of 1.00 versus 1.38; *P*=0.023), higher estimated glomerular filtration rate (90 versus 86 mL/min; *P*=0.036), and higher serum potassium (3.9 versus 3.7 mmol/L; *P*<0.001). They also had a lower prevalence of diabetes, microalbuminuria, and left ventricular hypertrophy (*P*=0.001 for all). Women

Table 2. Follow-Up Evaluation of Patients After Stratification for Biochemical Response

	Biochemical outcome			Overall *P* value	Pairwise comparisons		
Variable	Complete (n=653)	Partial (n=270)	Absent (n=197)		C vs P	C vs A	P vs A
Duration of FU, mo	36 (23–50)	31 (20–48)	34 (22–59)	0.062	…	…	…
SBP, mm Hg	130±14.2	134±14.1	139±18.6	<0.001*	0.002*	<0.001*	<0.001*
DBP, mm Hg	82±10.3	84±11.3	87±12.2	<0.001*	0.010*	<0.001*	0.024*
DDD	1.00 (0.00–2.15)	1.00 (0.00–2.00)	1.25 (0.33–3.00)	0.052	…	…	…
Use of βB (ref. yes), n (%)	126/653 (19.3)	64 (23.7)	57 (28.9)	0.013*	0.132	0.004*	0.203
Targeted treatment							
Spironolactone, mg/d	40.0 (25.0–50.0)	37.5 (25.0–50.0)	25.0 (20.0–50.0)	0.010*	1.000	0.007*	0.126
Eplerenone, mg/d	100.0 (50.0–100.0)	75.0 (50.0–100.0)	50.0 (50.0–100.0)	0.396	…	…	…
Esaxerenone, mg/d	3.8 (1.3–5.0)	5.0 (1.3–5.0)	5.0 (3.1–5.0)	0.529	…	…	…
Amiloride, mg/d	5 (5–10)	10 (5–10)	5.0 (5.0–10.0)	0.201	…	…	…
MRA dose (as DDD)	1.33 (0.67–2.00)	1.07 (0.67–2.00)	1.00 (0.53–1.50)	0.058	…	…	…
Aldosterone, pmol/L	797.5 (555.9–1160.6)	665.8 (471.5–1028.0)	619.5 (435.9–1041.9)	<0.001*	0.002*	0.002*	1.000
PRA, ng/mL per hour	2.71 (1.70–4.88)	0.63 (0.40–0.80)	0.32 (0.13–0.60)	<0.001*	<0.001*	<0.001*	0.872
DRC, mU/L	27.4 (16.0–52.3)	5.4 (3.7–7.7)	2.7 (1.9–5.3)	<0.001*	<0.001*	<0.001*	0.239
ARR (PRA)	252 (134–450)	961 (553–2369)	1477 (641–3048)	<0.001*	<0.001*	<0.001*	1.000
ARR (DRC)	30 (14–50)	122 (81–215)	227 (115–469)	<0.001*	<0.001*	<0.001*	0.074
K^+, mmol/L	4.4±0.40	4.3±0.40	4.0±0.55	<0.001*	0.001*	<0.001*	<0.001*
Creatinine, mg/dL	1.00±0.338	0.96±0.368	0.99±0.384	0.379	…	…	…
eGFR, mL/min	79±22.0	80±21.8	79±22.1	0.879	…	…	…
MA (ref. yes), n (%)	69/442 (15.6)	43/196 (21.9)	31/138 (22.5)	0.066	…	…	…
LVH at echo (ref. yes), n (%)	88/355 (24.8)	48/159 (30.2)	28/113 (24.8)	0.408	…	…	…
Delta SBP, mm Hg	−21 (−34 to −9)	−15 (−30 to −1)	−12 (−24 to +2)	<0.001*	0.001*	<0.001*	0.122
Delta DBP, mm Hg	−10 (−18 to +0)	−7 (−16 to +1)	−4 (−15 to +5)	<0.001*	0.053	<0.001*	0.157
Delta DDD	−0.83 (−2.00 to +0.00)	−0.60 (−1.50 to +0.00)	−0.25 (−1.17 to +0.50)	0.001*	0.615	0.001*	0.072
Delta PRA, ng/mL per hour	+2.25 (+1.20 to +4.19)	+0.30 (+0.20 to +0.60)	−0.02 (−0.18 to +0.00)	<0.001*	<0.001*	<0.001*	0.044*
Delta DRC, mU/L	+22.6 (+11.6 to +46.9)	+3.0 (+1.3 to +4.6)	−0.5 (−1.6 to +0.0)	<0.001*	<0.001*	<0.001*	<0.001*

Follow-up evaluation of patients after stratification for biochemical outcome according to PAMO criteria. ARR indicates aldosterone-to-renin ratio; DBP, diastolic blood pressure; βB, β-blockers; DDD, defined daily dose; DRC, direct renin concentration; echo, echocardiography; eGFR, estimated glomerular filtration rate; FU, follow-up; LVH, left ventricular hypertrophy; MA, microalbuminuria; PAMO, Primary Aldosteronism Medical treatment Outcomes; PRA, plasma renin activity; and SBP, systolic blood pressure.

*A *P*<0.05 was considered significant.

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Table 3. Baseline Characteristics of Patients After Stratification for Clinical Response

Variable	Clinical outcome			Overall *P* value	Pairwise comparisons		
	Complete (n=264)	Partial (n=811)	Absent (n=197)		C vs P	C vs A	P vs A
Age at referral, y	51±11.1	52±11.4	52±10.8	0.364	…	…	…
Duration of HTN, mo	24 (8–110)	72 (18–166)	96 (23–180)	<0.001*	<0.001*	<0.001*	0.538
Sex (ref. female), n (%)	178 (67.4)	347 (42.8)	82 (41.6)	<0.001*	<0.001*	<0.001*	0.767
BMI, kg/sqm	26.9±5.22	29.1±5.87	28.3±5.35	<0.001*	<0.001*	0.048*	0.153
SBP, mm Hg	144±18.0	154±20.1	144±15.4	<0.001*	<0.001*	1.000	<0.001*
DBP, mm Hg	89±11.8	93±13.0	88±10.5	<0.001*	<0.001*	1.000	<0.001*
DDD	1.00 (0.00–2.00)	2.24 (1.33–4.00)	1.38 (0.70–2.50)	<0.001*	<0.001*	0.023*	<0.001*
Use of βB (ref. yes), n (%)	20/262 (7.6)	189/784 (24.1)	26/191 (13.6)	<0.001*	<0.001*	0.037*	0.002*
AC, pmol/L	498.8 (344.0–679.2)	543.9 (398.5–777.8)	546.5 (423.7–770.8)	0.002*	0.002*	0.009*	1.000
PRA, ng/mL per hour	0.31 (0.20–0.59)	0.40 (0.20–0.60)	0.32 (0.20–0.53)	0.233	…	…	…
DRC, mU/L	2.7 (2.0–5.0)	3.3 (2.0–5.5)	3.6 (2.0–6.0)	0.053	…	…	…
ARR (PRA)	1657 (910–3204)	1425 (818–2915)	1719 (1041–3612)	0.200	…	…	…
ARR (DRC)	146 (97–251)	149 (92–259)	141 (92–246)	0.857	…	…	…
Creatinine, mg/dL	0.79±0.191	0.91±0.337	0.90±0.284	<0.001*	<0.001*	<0.001*	1.000
eGFR, mL/min	90±18.7	85±19.9	86±19.9	<0.001*	<0.001*	0.036*	1.000
K^+, mmol/L	3.9±0.42	3.8±0.49	3.7±0.46	<0.001*	0.002*	<0.001*	0.084
Lowest K^+, mmol/L	3.5±0.51	3.3±0.49	3.3±0.55	<0.001*	<0.001*	0.001*	1.000
Diabetes (ref. yes), n (%)	22 (8.3)	159 (19.6)	37 (18.8)	<0.001*	<0.001*	0.001*	0.793
MA (ref. yes), n (%)	14/185 (7.6)	141/623 (22.6)	32/166 (19.3)	<0.001*	<0.001*	0.001*	0.353
LVH at echo (ref. yes), n (%)	43/219 (19.6)	234/644 (36.3)	58/167 (34.7)	<0.001*	<0.001*	0.001*	0.700
CT scanning							
Nodule (ref. yes), n (%)	63/251 (25.1)	293/766 (38.3)	60/187 (32.1)	0.001*	<0.001*	0.108	0.118
Adrenal description, n (%)				<0.001*	<0.001*	0.002*	0.990
Bilaterally normal	135/241 (56.0)	298/738 (40.4)	75/187 (40.1)				
Bilaterally abnormal	22/241 (9.1)	127/738 (17.2)	33/187 (17.6)				
Unilateral abnormality	84/241 (34.9)	313/738 (42.4)	79/187 (42.3)				
Adrenal venous sampling without ACTH							
Lateralization index	1.8 (1.3–2.8)	2.1 (1.4–3.4)	2.1 (1.4–4.9)	0.221	…	…	…
CL suppr. (ref. yes), n (%)	38/97 (39.2)	121/332 (36.4)	37/86 (43.0)	0.518	…	…	…
Adrenal venous sampling with ACTH							
Lateralization index	1.6 (1.2–2.4)	1.6 (1.2–2.6)	1.6 (1.2–2.1)	0.240	…	…	…
CL suppr. (ref. yes), n (%)	23/109 (21.1)	71/337 (21.1)	13/74 (17.6)	0.787	…	…	…

Baseline patient characteristics after stratification for clinical outcome according to PAMO criteria. AC indicates aldosterone concentration; ACTH, adreno-corticotropic hormone; ARR, aldosterone-to-renin ratio; BMI, body mass index; CL, contralateral; CT, computed tomography; DBP, diastolic blood pressure; βB, β-blockers; DDD, defined daily dose; DRC, direct renin concentration; echo, echocardiography; eGFR, estimated glomerular filtration rate; HTN, hypertension; LVH, left ventricular hypertrophy; MA, microalbuminuria; PAMO, Primary Aldosteronism Medical treatment Outcomes; PRA, plasma renin activity; SBP, systolic blood pressure; and suppr., suppression.

*A $P<0.05$ was considered significant.

comprised a greater proportion of the complete response group compared with the absent response group (67.4% versus 41.6%; $P<0.001$). At follow-up (median 34–36 months), participants with a complete clinical response, compared with those with an absent clinical response, had a higher dose of MRA (DDD of 1.33 versus 1.00; $P=0.013$), higher renin concentration (17.0 versus 8.2 mU/L; $P=0.007$) and a greater increase in renin from baseline (12.6 versus 3.4 mU/L; $P<0.001$; Table 4).

In the regression analysis, a more favorable clinical response was independently associated with female sex (OR, 1.986 [1.432–2.755]; $P<0.001$), lower body mass index (OR, 0.960 [0.931–0.990]; $P=0.008$), lower duration of hypertension (OR, 0.978 [0.970–0.987]; $P=0.045$), and absence of known hypertension mediated organ damage (OR, 0.707 [0.505–0.989]; $P=0.043$), again considering geographic area, follow-up duration, and MRA dose as potential confounding factors (Table 5).

Patterns of MRA titration differed across biochemical and clinical response categories, with fewer patients undergoing MRA up-titration among those with an absent response (Table S6). Consistent with this, in the regression analysis including only patients receiving spironolactone, a higher DDD was independently

Table 4. Follow-Up Evaluation of Patients After Stratification for Clinical Response

Variable	Clinical outcome			Overall *P* value	Pairwise comparisons		
	Complete (n=264)	Partial (n=811)	Absent (n=197)		C vs P	C vs A	P vs A
Duration of FU, mo	34 (21–52)	34 (21–51)	36 (22–53)	0.919	…	…	…
SBP, mm Hg	124±9.3	132±14.3	144±18.9	<0.001*	<0.001*	<0.001*	<0.001*
DBP, mm Hg	78±7.1	82±10.8	91±12.1	<0.001*	<0.001*	<0.001*	<0.001*
DDD	0.00 (0.00–0.00)	1.50 (1.00–3.00)	2.00 (1.00–3.50)	<0.001*	<0.001*	<0.001*	0.087
Use of βB (ref. yes), n (%)	0 (0.0)	238 (29.3)	45 (22.8)	<0.001*	<0.001*	<0.001*	0.068
Targeted treatment							
Spironolactone, mg/d	40.0 (25.0–50.0)	40.0 (25.0–50.0)	25.0 (20.0–50.0)	0.085	…	…	…
Eplerenone, mg/d	100.0 (50.0–100.0)	75.0 (50.0–100.0)	50.0 (50.0–100.0)	0.021*	0.694	0.022*	0.073
Esaxerenone, mg/d	5.0 (2.2–5.0)	5.0 (1.3–5.0)	4.5 (2.2–5.0)	0.868	…	…	…
Amiloride, mg/d	10.0 (10.0–10.0)	5.0 (5.0–10.0)	5.0 (5.0–10.0)	0.749	…	…	…
MRA dose (as DDD)	1.33 (0.67–2.00)	1.17 (0.67–2.00)	1.00 (0.53–1.33)	0.007*	1.000	0.013*	0.010*
AC, pmol/L	710.1 (473.0–1054.1)	749.0 (504.1–1108.8)	764.0 (542.1–1123.9)	0.341	…	…	…
PRA, ng/mL per hour	1.32 (0.50–3.10)	1.53 (0.70–3.69)	1.12 (0.40–2.58)	0.109	…	…	…
DRC, mU/L	17.0 (7.0–29.9)	15.9 (6.1–38.6)	8.2 (3.7–27.0)	0.001*	1.000	0.007*	0.001*
ARR (PRA)	497 (248–1123)	420 (194–881)	529 (224–1367)	0.265	…	…	…
ARR (DRC)	58 (32–111)	48 (19–120)	79 (28–208)	0.011*	0.884	0.280	0.009*
K^+, mmol/L	4.3±0.41	4.3±0.45	4.2±0.48	<0.001*	1.000	0.005*	<0.001*
Creatinine, mg/dL	0.88±0.222	1.03±0.392	1.00±0.373	<0.001*	<0.001*	0.003*	0.934
eGFR, mL/min	82±20.7	77±22.4	80±21.0	0.003*	0.003*	0.741	0.375
MA (ref. yes), n (%)	10/159 (6.3)	110/534 (20.6)	33/152 (21.7)	<0.001*	<0.001*	<0.001*	0.766
LVH at echo (ref. yes), n (%)	16/128 (12.5)	125/408 (30.6)	33/124 (26.6)	<0.001*	<0.001*	0.005*	0.390
Delta SBP, mm Hg	−19 (−35 to −7)	−22 (−34 to −9)	−1 (−11 to +8)	<0.001*	0.585	<0.001*	<0.001*
Delta DBP, mm Hg	−11 (−20 to −1)	−11 (−20 to −2)	+2 (−4 to +10)	<0.001*	1.000	<0.001*	<0.001*
Delta DDD	−1.00 (−2.00 to +0.00)	−0.99 (−2.00 to +0.16)	+0.30 (+0.00 to +1.10)	<0.001*	<0.001*	<0.001*	<0.001*
Delta PRA, ng/mL per hour	+0.96 (+0.20 to +2.50)	+1.20 (+0.30 to +3.27)	+0.72 (+0.00 to +2.11)	0.205	…	…	…
Delta DRC, mU/L	+12.6 (+4.3 to +26.2)	+10.8 (+2.5 to +33.8)	+3.4 (+0.1 to +20.7)	<0.001*	1.000	<0.001*	<0.001*

Follow-up evaluation of patients after stratification for clinical outcome according to PAMO criteria. AC indicates aldosterone concentration; ACTH, adreno-corticotropic hormone; ARR, aldosterone-to-renin ratio; DBP, diastolic blood pressure; βB, β-blockers; DDD, defined daily dose; DRC, direct renin concentration; echo, echocardiography; eGFR, estimated glomerular filtration rate; FU, follow-up; LVH, left ventricular hypertrophy; MA, microalbuminuria; PAMO, Primary Aldosteronism Medical treatment Outcomes; PRA, plasma renin activity; and SBP, systolic blood pressure.

*A $P<0.05$ was considered significant.

associated with a favorable biochemical response (OR, 1.634 [1.189–2.245]; P=0.002; Table S7).

Finally, although follow-up duration did not differ significantly across biochemical and clinical response groups, its spread at outcome assessment was relatively wide, varying from 12 to 156 months; the broad range of follow-up duration, even if it offered a unique perspective on sustained biochemical and clinical outcomes that cannot be captured in shorter studies, it also introduces heterogeneity; for this reason, we stratified data on patients' outcomes by follow-up quartiles (Table S8). The rate of complete biochemical response progressively increased with longer follow-up duration (from 52.2%–66.5%; P=0.025), whereas the clinical response rate remained unchanged. No significant differences in MRA dose were observed across quartiles, although DDD was numerically higher in the group with the longest follow-up.

DISCUSSION

In this international cohort study of 1292 patients with PA treated with targeted medical therapy, we provide the first assessment of sustained clinical and biochemical outcomes, as well as the factors independently associated with a more favorable response, at 12 months and beyond. Our findings build on prior work examining treatment response at 6 to 12 months and provide new insights into the durability and determinants of longer-term medical treatment response in real-world settings.

A key observation was the relatively high proportion of patients achieving favorable biochemical outcomes over a median follow-up of 34 months, with the proportion being higher with a longer duration of follow-up. Considering participants with paired follow-up data at both 6 to 12 months and ≥12 months, 60.8% achieved a complete

Table 5. Determinants of Biochemical and Clinical Response

	OR	95% CI Low limit	95% CI High limit	*P* value
Biochemical outcome				
Age, y	1.013	0.997	1.028	0.108
Female sex	0.979	0.716	1.339	0.896
BMI, kg/sqm	1.008	0.979	1.038	0.596
Baseline aldosterone, pmol/L	0.999	0.998	0.999	0.035*
Baseline DRC, mU/L†	1.097	1.045	1.152	<0.001*
Baseline K^+, mmol/L	1.543	1.065	2.233	0.022*
Bilaterally normal adrenals	1.021	0.651	1.600	0.929
LI >4 at AVS (with/without ACTH)	0.666	0.448	0.989	0.044*
Geographic area (ref. America)				
Asia	0.810	0.418	1.571	0.534
Europe	0.548	0.300	1.010	0.060
Oceania	0.491	0.209	1.009	0.058
Follow-up duration, mo	0.996	0.989	1.004	0.352
MRA dose (DDD)	1.162	0.970	1.391	0.104
Use of βB (at follow-up)	0.531	0.361	0.782	0.001*
Clinical outcome				
Age, y	0.999	0.983	1.016	0.911
Female sex	1.986	1.432	2.755	<0.001*
BMI, kg/sqm	0.960	0.931	0.990	0.008*
Duration of HTN, mo	0.978	0.970	0.987	0.045
Baseline SBP, mm Hg	1.000	0.992	1.008	0.933
Baseline DDD	0.996	0.909	1.091	0.930
Bilaterally normal adrenals	0.710	0.451	1.118	0.139
LI >4 at AVS (with/without ACTH)	0.683	0.447	1.043	0.078
Known HMOD (MA and LVH)	0.707	0.505	0.989	0.043*
Geographic area (ref. America)				
Asia	0.699	0.352	1.391	0.308
Europe	0.744	0.397	1.397	0.358
Oceania	1.198	0.548	2.617	0.651
Follow-up duration, mo	1.003	0.995	1.011	0.430
MRA dose (DDD)	1.125	0.913	1.387	0.267

Three-level proportional odds ordinal logistic regression models to identify parameters associated with biochemical or clinical outcomes. Patients' responses to treatment were treated as ordered categories (complete > partial > absent); adjusted OR with 95% CI represent the likelihood of being in a more favorable outcome category. Covariates included in the model and potential confounding factors (below the line: geographic area, follow-up duration, MRA dose, and use of β-blockers) were prespecified based on clinical relevance and previous evidence. OR for geographic area are reported using America as the reference category (OR represents the likelihood of being in a more favorable category for each of the following comparisons: Asia vs America, Europe vs America, and Oceania vs America). ACTH, adreno-cortico-tropic hormone; AVS, adrenal vein sampling; BMI, body mass index; DBP, diastolic blood pressure; DDD, defined daily dose; βB, β-blockers; DRC, direct renin concentration; HMOD, hypertension-mediated organ damage; HTN, hypertension; LI, lateralization index; LVH, left ventricular hypertrophy; MA, microalbuminuria; MRA, mineralocorticoid receptor antagonist; OR, odds ratio; PRA, plasma renin activity; and SBP, systolic blood pressure.

†PRA levels were converted to DRC by a conversion factor of 10, according to the following formula: DRC (mU/L)=PRA (ng/mL per hour)×10.

biochemical response, while 22.3% achieved a complete clinical response at the later time point compared with 54.1% and 18.9%, respectively, at the earlier time point.[8] The median doses of spironolactone and eplerenone in patients with a complete (40 and 100 mg, respectively) or absent (25 and 50 mg, respectively) biochemical response were comparable at the 2 time points, while the median doses were higher at the later time point in the partial response group (spironolactone 31 mg at 6–12 months, 37.5 mg at ≥12 months; eplerenone 50 mg at 6–12 months, 75 mg at ≥12 months). The highest rate of complete biochemical response at 66.5% was observed in the quartile with the longest follow-up (52–156 months); this quartile also had the highest median MRA dose. These response rates suggest that the therapeutic benefit of targeted treatment is durable and may continue to accrue over time with appropriate dose titration. The relatively low proportion of patients achieving a complete clinical response may be disappointing, but it is important to note that blood pressure reduction in the complete and partial clinical response groups was comparable at 19 and 22 mm Hg for systolic and 11 mm Hg for diastolic blood pressures. The reduction in the DDD of additional antihypertensives was also similar between these groups, at both 6 to 12 months and ≥12 months. Our data showed that partial clinical responders had a higher prevalence of comorbidities such as diabetes, microalbuminuria, and left ventricular hypertrophy, which probably warranted continued treatment with ACE (angiotensin-converting enzyme) inhibitors, angiotensin II receptors, diuretics, or other antihypertensive medications that placed them in the partial response category. The data strongly support the value of targeted medical therapy for patients with PA, even when the clinical response is categorized as partial.

An evaluation of the concordance between biochemical and clinical responses demonstrated that these represent related but distinct domains of treatment success. Although a statistically significant association was observed between response categories, concordance was poor, indicating that achieving biochemical control does not reliably translate to complete clinical response. This dissociation likely reflects that blood pressure response is multifactorial and influenced by other factors such as disease duration, vascular remodeling, and comorbidities, meaning additional antihypertensive therapy may be required even after achieving a complete biochemical response. These findings suggest that biochemical and clinical outcomes should be considered complementary rather than interchangeable targets, and that treatment optimization in PA requires attention to both domains.

Among those with a complete biochemical or clinical response, the median dose of spironolactone was 40 mg daily, compared with 25 mg daily in patients with an absent response. When spironolactone users were analyzed separately, a higher DDD was significantly associated with a complete biochemical response. The lower proportion of patients who underwent MRA up-titration from 6 to 12 months to ≥12 months in the absence of biochemical response group is notable. Given that this group had similar age, sex distribution, serum potassium, and estimated glomerular filtration rate to the complete and partial biochemical response groups, this finding may reflect other dose-limiting adverse effects that have not been captured by our study, or a lack of awareness regarding the need to up-titrate MRA therapy in the context of persistently low renin and elevated blood pressure. Our study included patients treated between 2016 and 2021, a period that coincided with emerging evidence supporting renin normalization as a therapeutic target,[13] but preceded any formal recommendations.[5,8,14] As such, clinicians may have prioritized adjustment of other antihypertensive agents rather than further increasing the MRA dose.

Aside from the dose of targeted therapy, the timeliness of disease detection also influences treatment response. We found that features indicative of earlier-stage disease, such as shorter duration of hypertension and absence of known hypertension mediated organ damage, are associated with a more favorable clinical response. These characteristics likely reflect a milder PA phenotype or a lower degree of aldosterone-induced vascular and kidney injury, making reversal more achievable with medical therapy. These observations support new guideline recommendations for early screening and prompt initiation of targeted treatment in patients with PA, as there may be a higher chance of full clinical response.

The subtype of PA can influence treatment response as well. Patients with bilateral subtypes of PA were more likely to achieve a complete biochemical response than those with unilateral disease (59.5% versus 49.3%), with fewer showing an absent response (16.5% versus 29.2%). In addition, a higher baseline plasma aldosterone concentration and a lateralization index >4 on adrenal vein sampling were associated with a lower likelihood of achieving complete biochemical response. This difference is likely attributable to a greater aldosterone burden in unilateral PA, as indicated by higher aldosterone concentrations at baseline. This finding aligns with a retrospective study of 155 patients on long-term medical therapy for PA, which reported a higher average spironolactone dose for patients with unilateral PA than bilateral PA (57 versus 50 mg).[15] Despite this, both groups received comparable doses of MRA in the current study. This finding again highlights the need for MRA dose escalation when biochemical and clinical responses are not optimal, especially in patients with unilateral disease who require higher doses of MRA to achieve sufficient renin normalization and biochemical control.

A notable finding was the significantly higher likelihood of complete clinical response in women, even though sex had no effect on the biochemical response. This association may be driven by several biological factors. Estrogen is known to positively influence vascular tone and endothelial function, potentially enhancing responsiveness to aldosterone blockade,[16] while progesterone can directly block the mineralocorticoid receptor.[17] Women in this cohort had a mean age of 50 years (SD, 11.1 years), indicating that many were likely premenopausal and may have benefited from estrogen-mediated cardiovascular protection. Additionally, more women used spironolactone than men, which is expected due to the anti-androgenic side effects in men. While spironolactone is more potent than eplerenone, the second most commonly used MRA, this difference does not explain the observed discordance between comparable biochemical response and superior clinical response in women. Longitudinal follow-up is warranted to determine whether this clinical advantage is sustained with aging and the transition through menopause.

This study has several strengths. It represents the largest international cohort to date evaluating medical treatment outcomes in patients with PA, encompassing 1292 patients across 27 centers and 4 continents. Such a scale was necessary because there is substantial variation in disease severity, treatment practices, and follow-up across centers; smaller or single-center studies lack sufficient power and diversity to assess the durability of treatment response or identify predictors of sustained benefit. This large and diverse sample enhances generalizability and provides insights into real-world clinical practice. The use of the PAMO criteria allowed for standardized classification of treatment response, enabling meaningful comparison across sites.

However, several limitations should be acknowledged. The observational and retrospective design introduces the potential for selection bias and unmeasured confounding, such as differences in salt intake, which can affect both renin and blood pressure. Although standard definitions were applied centrally, data collection was site-reported and may be subject to variation in clinical practice, measurement methods or follow-up protocols. For example, the use of interfering medications such as diuretics and renin-angiotensin-system modulators could have increased renin concentration, while β-blockers could have decreased renin concentration, and caused misclassification of the biochemical response. Similarly, tolerability of and adherence to MRA therapy could also impact the classification of treatment response, but the relevant data were not collected in this study. In real practice, blood pressure and renin normalization can still serve as treatment targets even if there are interfering medications or reasons for limiting dose escalation. However, the clinician should account for individual variations rather than just aiming for complete response categorization because it may not be feasible to achieve a complete biochemical or clinical response. Another limitation is the lack of data on epithelial sodium channel inhibitors such as amiloride, which was taken by only 43 patients. Hence, treatment outcomes for this class of drugs remain to be defined. Additionally, while the PAMO criteria provide a structured framework for evaluating treatment response, they do not capture patient-reported outcomes or long-term cardiovascular events, which is a key consideration that underpins the recommendation to normalize renin as a treatment target.[14,18] There are currently no data on the impact of renin normalization in the context of controlled blood pressure. Further evaluation of major cardiovascular end points in relation to PAMO categories, as well as analysis of MRA up-titration according to blood pressure or renin in prospective trials are warranted.

Finally, although this was a real-world study, the data were mostly drawn from tertiary centers with dedicated expertise in PA management. Hence, the actual rates of biochemical and clinical control in less specialized centers may be lower than what has been reported in this study.

Conclusions

Our findings demonstrate that a substantial proportion of patients with PA can achieve sustained biochemical and clinical improvement beyond 12 months of treatment with appropriately dosed targeted medical therapy. However, clinical response remains more limited than biochemical response, particularly in patients with more severe disease at baseline. These results suggest that early disease detection and timely initiation of targeted treatment with adequate dose titration may lead to more complete biochemical and clinical response over time. By demonstrating durable treatment response and defining determinants of sustained benefit across diverse populations, this study provides critical evidence to guide long-term medical management of PA. It also highlights opportunities to optimize care delivery and reduce international disparities in the management of PA.

Perspectives

PA is increasingly recognized as a common and treatable cause of hypertension, yet there has been limited data regarding the long-term effectiveness of medical therapy and reliable determinants of optimal treatment responses. The PAMO beyond 12 months study addresses this gap by providing the largest and most geographically diverse evaluation of long-term medical treatment outcomes in PA to date.

By extending follow-up to a median of nearly 3 years across 27 centers on 4 continents, this study demonstrates that targeted medical therapy can deliver sustained biochemical control in the majority of patients, with meaningful clinical improvement in terms of improved blood pressure control and reduced requirement for additional antihypertensives. The identification of patient characteristics associated with favorable long-term outcomes, such as a shorter duration of hypertension and absence of hypertension mediated organ damage at baseline, further supports the value of early disease detection and intervention before irreversible end-organ damage occurs. Spironolactone dose was found to be associated with complete biochemical response, emphasizing the importance of adequate dose titration in the medical management of PA.

These findings provide a framework for optimizing medical management in PA and support the integration of PAMO criteria into longitudinal care pathways. Future prospective studies linking PAMO response categories with cardiovascular and patient-reported outcomes will be important for defining the long-term benefits of targeted therapy in PA.

ARTICLE INFORMATION

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